

SAR Trust Study

Pilot Analysis Report — Session 1 Snapshots

Two 7-minute pilots; one per complexity mode. Both used *mode=agent*, *condition=none*, $\epsilon=0$ (perfect accuracy). Snapshots captured mid-session; sessions 2 and 3 untouched.

Metric	P-3666 (tactical)	P-2673 (strategic)
Session elapsed	383s (~6.4 min)	329s (~5.5 min)
Penalty accrued	535	150
Missions arrived	13	10
Missions completed	0	2
Missions abandoned	5	2
Strategic: agent followed	5/6 (83%)	2/7 (28%)
Tactical: agent modified	5/5 (100%)	2/2 (100%)
Drone failures (total)	9	7
Blue drones failed	8	6

Part 1: Findings

1.1 Mission Outcomes

P-3666 (tactical) completed **zero missions** in 6.4 minutes and accrued 535 penalty points — 3.6× the penalty of P-2673 (strategic, 150 pts). Five missions were abandoned mid-execution; seven more sat queued and never reached allocation. P-2673 completed two missions (M001 and M003) with two abandoned, three still active, and three queued at the snapshot.

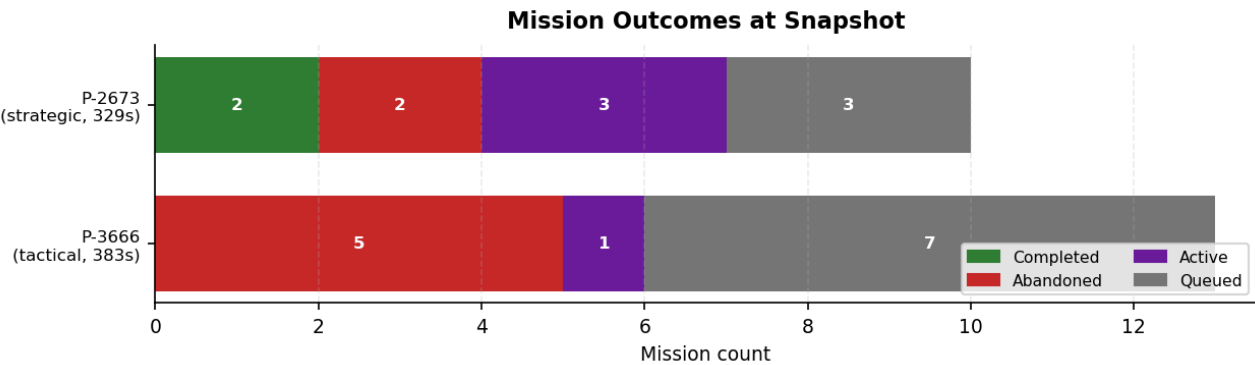


Figure 1. Mission status at snapshot. Numbers inside bars are mission counts.

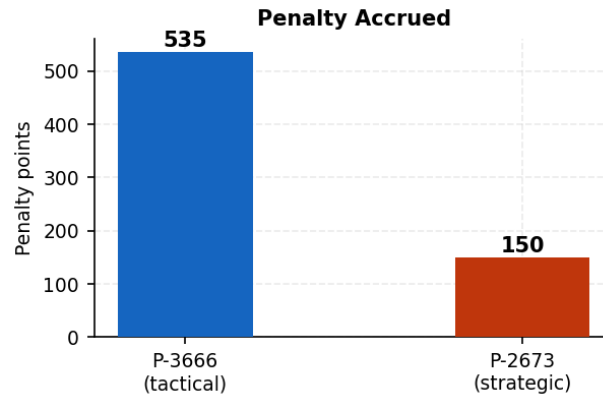


Figure 2. Penalty accrued. P-3666 (tactical) was 3.6× worse despite following the agent more often, pointing to a difficulty imbalance rather than a behavioural one.

The tactical complexity scenario had missions with 5–6 tasks and very high drone failure density. By elapsed 250 s (under halfway), all ten Blue drones were destroyed or failed in P-3666, leaving the operator unable to staff subsequent missions even when ready to allocate.

1.2 Strategic Agent Follow Rate

P-3666 followed the Strategic Agent in **5 of 6 allocated missions (83%)**. P-2673 followed it in only **2 of 7 (29%)**, preferring manual allocation for five missions including the first three in the session. The divergence is striking given that both runs used $\epsilon=0$ (perfectly accurate agent). It is not an accuracy effect — it is likely a UI familiarity or trust-calibration effect that will need watching in the real study.

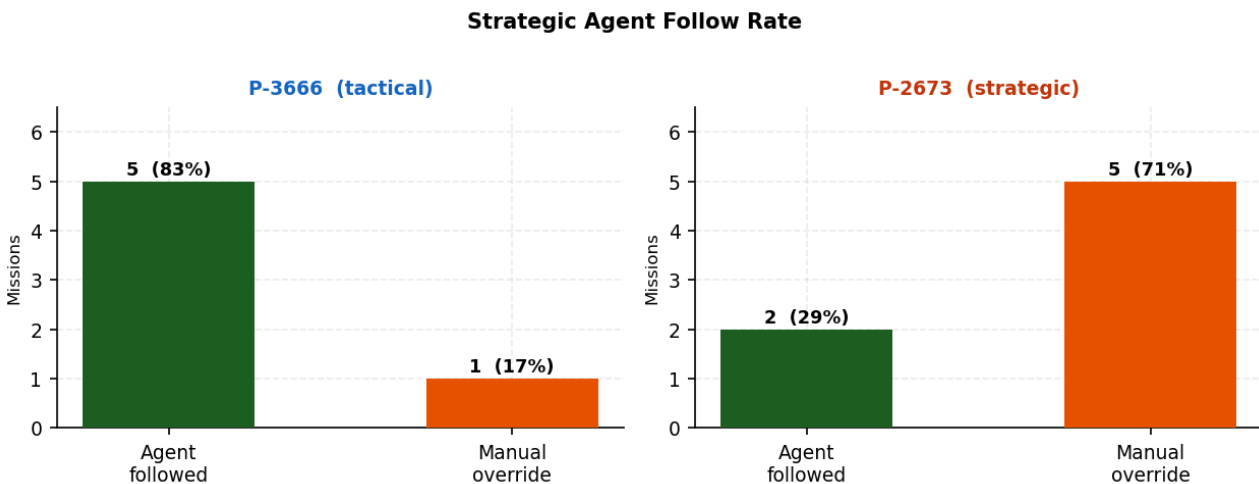


Figure 3. Final strategic choice per mission: agent-followed vs. manual. Percentages are of total allocated missions.

1.3 Tactical Agent Modification Rate

In both pilots, the operator modified the Tactical Agent's assignment plan **100% of the time** it was presented (5/5 in P-3666; 2/2 in P-2673). The plan was never accepted without drag-and-drop changes. Manual allocations (where no agent plan was shown) were always confirmed as-is.



Figure 4. Tactical confirmations by category. "Agent-suggested & accepted unmodified" is zero for both pilots.

This does not necessarily mean the plan was wrong — operators may have had legitimate reasons (drone position, personal strategy). However a 100% modification rate warrants investigation before the live study.

1.4 Decision Latency

Latency is measured from the last strategic modal open to the final strategic choice for each mission. Early missions in both runs were decided quickly (3–8 s). Later missions took substantially longer: up to 24 s for P-3666's M006, and P-2673's M001 remained in the modal workflow for over 35 s across three sequential opens.

Decision Latency (last modal open → choice)

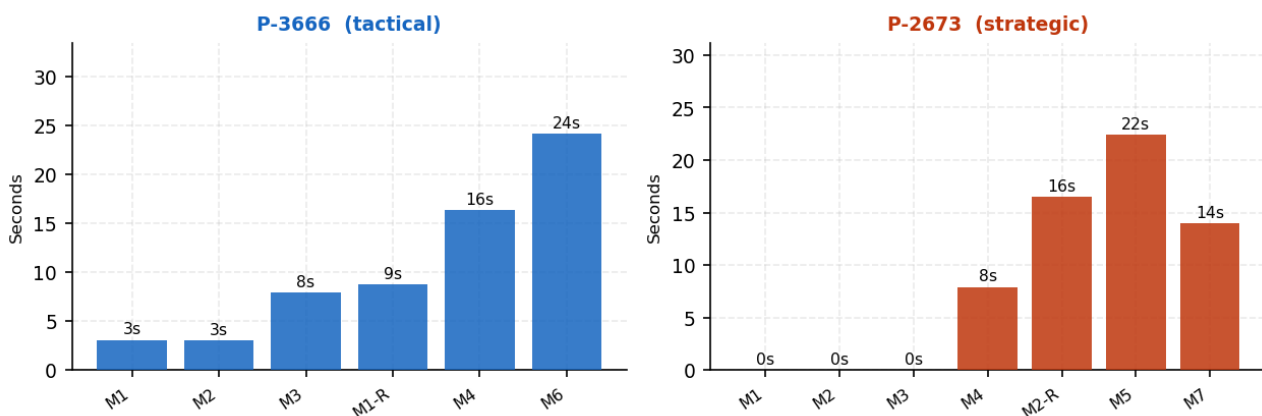


Figure 5. Decision latency per mission (last modal open → choice). Missions in chronological order left to right.

1.5 Strategic Modal Re-opens

Re-opening the modal (count > 1) is a proxy for hesitation or dissatisfaction with the initial options. P-2673's M001 was opened **three times** — the operator made two manual choices before finally deploying. Three missions across each pilot had their modals re-opened at least once. Two missions (M008 in P-3666; M008 in P-2673) had the modal open with an **empty strategy list** — the system opened the dialog but had no valid suggestions to show due to depleted reserves.

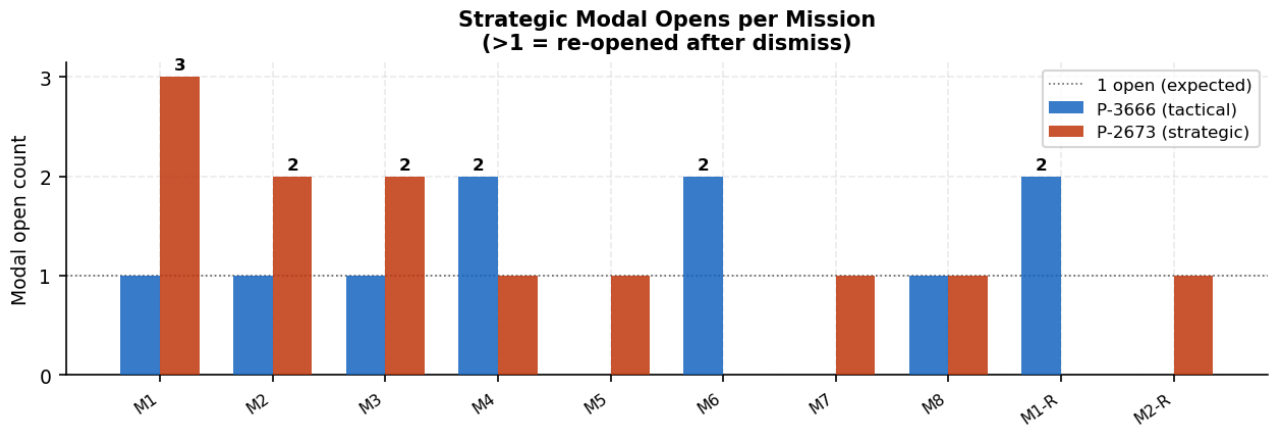


Figure 6. Strategic modal open count per mission. Dotted line = 1 (expected single open). Missions not present in a pilot had no modal event.

1.6 Drone Failure Timeline

Both pilots experienced similar overall failure rates (~1.3–1.4 per minute). The key difference is *which* drones failed. In P-3666 (tactical), failures concentrated in Blue drones — 8 of 9 total failures were Blues — because tactical missions committed Blue drones to high-attribution Type-5 tasks. By ~250 s the entire Blue fleet was gone. P-2673's failures were distributed more evenly (5 Blue, 1 Red) and the fleet remained partially functional at snapshot.

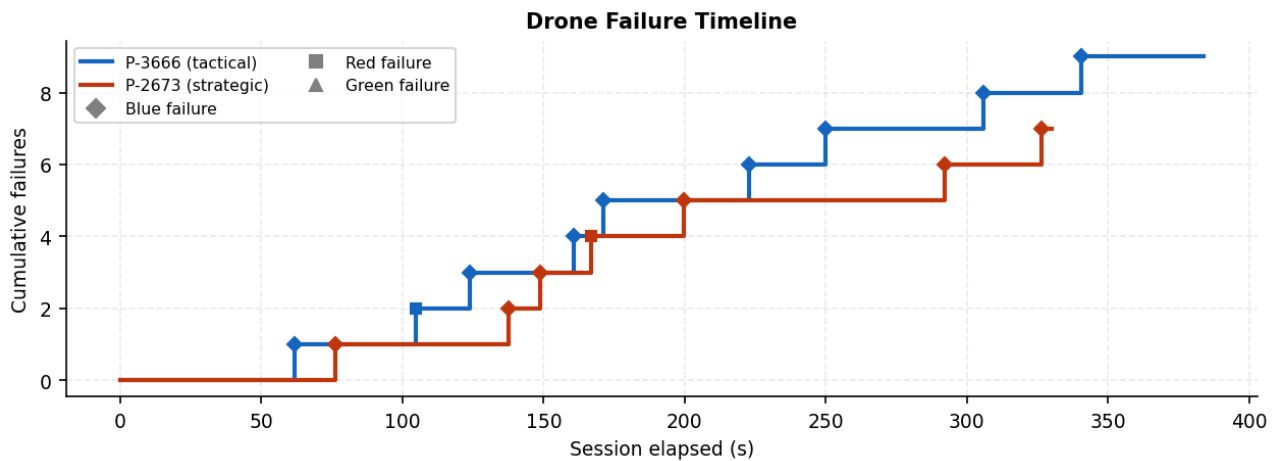


Figure 7. Cumulative drone failures over time. Markers show drone type: ◆ Blue, ■ Red, ▲ Green.

Asset Health at Snapshot

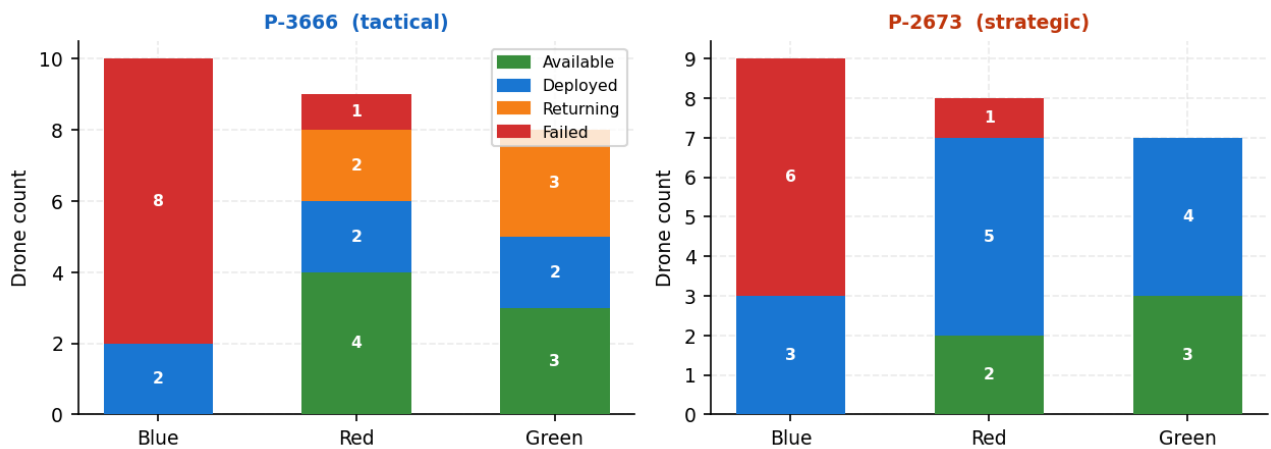


Figure 8. Asset health at snapshot. Colours: green=available, blue=deployed, orange=returning, red=failed.

Part 2: What Needs to Change

These findings come from one pilot per mode with a single operator under no-noise conditions ($\epsilon=0$). Confidence is limited, but several issues are structural enough to address before the live study.

2.1 Difficulty Balance — Tactical Complexity Is Too Hard

Issue: P-3666 completed zero missions in 6+ minutes and had its entire Blue fleet wiped out by the halfway point. With zero positive feedback events and a compounding penalty, a real participant would likely disengage or attribute performance failure to the interface rather than their own strategy.

- Check whether the mission generator is correctly applying the "tactical" complexity parameters. If task counts and Poisson arrival rate are both set to hard levels simultaneously, the combination may be unintentional.
- Consider decoupling *mission complexity* (task count, drone commitment) from *session difficulty* (arrival rate). The study needs cognitive load variance, not a death spiral.
- Add a floor on reserve: if the reserve drops below a threshold, pause new arrivals briefly or cap the strategic agent's recommended commitment. Operators should not be able to commit their last Green drone.
- Run a calibration pass: aim for ≥ 1 completion per 7-minute window in both complexity modes so participants experience at least some success.

! This must be resolved before any condition-based between-subjects comparison is meaningful — difficulty differences will swamp condition effects.

2.2 Empty Strategy Modal Must Not Fire

Issue: When `strategiesPresented` is empty the modal still opens, presenting the operator with a blank dialog. This happened on M001-R (first open) and M008 in P-3666, and M008 in P-2673. In all cases no allocation followed.

- Gate the modal on `strategiesPresented.length > 0`. If no strategy can be generated, either skip straight to the tactical planner with a "manual allocation required" message, or show an inline notice in the mission queue.
- Log the "no viable strategy" event explicitly so it can be tracked in analysis.

2.3 Tactical Plan — Investigate Why 100% Are Modified

Issue: Every tactical plan presented by the agent was edited before confirmation. This is concerning because the tactical follow-rate is one of the study's dependent variables — if it always floors at 0% regardless of condition, it cannot discriminate.

- Audit `greedyAssign()` output against what was actually deployed. Check whether the suggested asset IDs differ from what the operator chose in obvious ways (e.g. agent picks B01 which is about to fail, operator manually swaps to a healthy one).
- Check whether the *display* of the tactical plan is clear. If operators do not recognise the agent's suggestion as a recommendation they should evaluate, they may edit by default as a "confirmation" action.
- Consider logging a finer-grained diff: which specific task assignments were changed vs. which were kept. "Modified" is a binary flag but the plan might be 90% followed with one swap.
- If operators are drag-dropping tasks because chaining is confusing (not because they disagree with the plan), the chaining UI may need a clearer visual explanation.

2.4 Strategic Modal Re-open Flow

Issue: M001 in P-2673 was opened three times with two intermediate manual choices — suggesting the operator was unsure how to adjust a custom allocation after committing, or inadvertently closed the modal.

- Add a "Change allocation" button to the active mission panel so operators can revise without fully re-opening the strategic modal from scratch.
- If the modal is dismissed without a choice, warn the operator that the mission remains unallocated, rather than silently closing.
- Consider whether the manual asset-count entry UX is clear: P-2673 tried three different custom bundles (1B1R1G, then 2B, then 2B confirmed). The controls may not give enough feedback about viability.

2.5 Low Agent Follow Rate in Strategic Complexity Mode

Issue: P-2673 went manual on 5/7 strategic decisions despite a perfectly accurate agent ($\epsilon=0$). Smaller missions (2–4 tasks, Blue-only) may make the agent's suggestions look over-engineered relative to "just send 2 Blues".

- Verify the agent generates sensible cards for small missions. For M001 (2 tasks, Blue-only) the Conservative card recommended 4B+1R+1G — the operator deployed 2 Blues and completed it fine. The suggestion may have looked wasteful.
- Add a third piece of information to strategy cards: *minimum viable bundle* vs. agent recommendation, so operators can calibrate how conservative the agent is being.
- Consider whether the strategic-mode complexity missions should include more mixed-type tasks so the agent's multi-drone orchestration is visibly necessary, not optional.

2.6 Logging Gaps

Analysis was limited to what the snapshot captures. Some events were not available:

- **No failure_recovery events logged** — M005 and M007 in P-2673 had `failureRecoveryPending: true` at snapshot but no recovery actions appear in the event stream. Confirm this event fires correctly.
- **No task_failed events** — abandoned missions still had tasks in "executing" or "traveling" state at abandonment. It is unclear whether these fire a `task_failed` event.
- **Tactical modification granularity** — as noted above, the binary `modifiedFromAgentPlan` flag should be supplemented by a per-task diff for richer analysis of where operator trust breaks down.

Report generated automatically from mid-session snapshot logs. P-3666 snapshot at 384 s elapsed, P-2673 at 330 s elapsed. Both runs: `mode=agent, condition=none,  $\epsilon_S=\epsilon_T=0$` .